system #2

$$\begin{cases} 1\cdot x+\frac{6}{4}\cdot y=1 & A_2=\frac{1}{4}\cdot A_1 \\ 3\cdot x+7\cdot y=13 & B_2=B_1 \end{cases}$$

system #3

$$\begin{cases} 1\cdot x+\frac{6}{4}\cdot y=1 & A_3=A_2 \\ 0\cdot x+\frac{10}{4}\cdot y=10 & B_3=-3\cdot A_2+B_2 \end{cases}$$

comes from $-3\cdot(1\cdot x)+3\cdot x$

comes from $-3\cdot\left(\frac{6}{4}\cdot y\right)+7\cdot y$

comes from $-3\cdot(1)+13$